Language Learning to Fluency: Analysing the Factors Associated with Learner Dropout

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1. Introduction

Language-learning platforms invest heavily in acquiring and retaining learners. This analysis investigates which learner behaviours and characteristics are associated with dropout.

This report examines language learner habits and their association with dropout rates, using a dataset of 45,000 learners. The dataset contains 18 variables such as age, study hours, and consistency.

Key Findings

- 37% of learners dropped out before B2; 63% did not.
- Total study hours (r = −0.29), comprehensible input hours (r = −0.27) and language difficulty (r = +0.27) showed the strongest associations with dropout.
- Dropout fell from 53% to 19% between the lowest and highest study-hour groups, and from 48% to 25% between the lowest and highest motivation groups.
- Prior languages known and age started showed almost no association.
- Learners with high study hours, active use and input dropped out at 10.5% versus 54.3% for those low on all three.

2. Data

2.1 Dataset overview

The Language Learning to Fluency dataset contains 45,000 learner records and 18 variables. It is a synthetic dataset published on Kaggle (“Language Learning to Fluency: Who Gets There?”, Sergio Nefedov) and does not contain real learner records. The data is clean and well structured, with no missing values or duplicate rows, making it suitable for analysis.

2.2 Variable Structure

Variable	Type	Description
learner_id	string	Unique learner ID
fsi_category	int	Foreign Service Institute (**FSI**) language categories, based on the number of class hours needed to reach proficiency. **Category 1**: “Easy” (e.g. Spanish), **Category 4**: “Exceptionally difficult” (e.g. Mandarin/Arabic)
related_language	int	0: The learner does not know a related language to target language (TL) 1: The learner knows a similar language to TL.
prior_languages	int	Total of languages already spoken.
age_started	int	Age when the learner began to study TL.
total_study_hours	int	Total study hours dedicated to TL.
active_use_share	float	Proportion of study time spent actively using the language (e.g. speaking, writing or conversing), rather than passive study. Ranges from 0 to 1.
immersion_months	float	Months spent living/using the language immersively
speaking_practice	float	Speaking/output practice (0-10)
feedback_quality	float	Corrective feedback quality, e.g. a tutor (0-10)
comprehensible_input_hours	int	Hours dedicated to comprehensible input (listening/reading)
uses_srs	int	0: Does not use spaced repetition 1: Uses spaced repetition
motivation	float	Motivation rated 0 to 10
consistency	float	Consistency rated 0 to 10
accent_nativelike	float	How native-like the accent is (0-10)
cefr_level	string	Attained CEFR level (A0 to C2)
reached_fluency	int	0: Did not reach CEFR level B2 1: Reached B2 or above
dropped_out	int	0: Did not drop out before reaching B2 1: Quit before reaching B2

2.3 Data Preparation

Data Loading: The dataset was imported into Python using Pandas.
Data Exploration: First, df.head() was used to view the content and structure of the first 5 rows of the dataset. Then, df.info() was used to view the variables, data types, number of rows and columns, and identify any null values. df.describe() was then used to obtain summary statistics for the numerical variables.
Data Quality Assessment: Missing values were assessed using df.isnull().sum(), none were found. Duplicate records were identified using df.duplicated().sum(), no duplicate rows were found in the dataset.

3. Methods

Data was cleaned and explored in Python (using pandas and matplotlib), loaded into PostgreSQL, and queried in SQL (PostgreSQL). SQL was utilised to study dropout rates by FSI category and by combinations of study hours, active use and comprehensible input. Python (pandas and matplotlib) was used to examine each factor in this report, including retention, group comparisons, correlations and figures. For each factor, mean values were first compared between learners who dropped out and those who did not. Continuous variables, such as motivation, consistency and study hours, were then grouped into categories (e.g. Low to High) so that dropout rates could be compared across levels and the gradient of change shown. Pearson correlations between each numeric variable and dropout were used to rank the factors by strength of association.

Data was cleaned and explored in Python (pandas and matplotlib), then loaded into PostgreSQL for querying in SQL. SQL was used to calculate dropout rates by FSI category and by combinations of study hours, active use and comprehensible input; Python was used for all other analysis, including retention, group comparisons, correlations and figures.

For each factor, mean values were first compared between learners who dropped out and those who did not. Continuous variables, such as motivation, consistency and study hours, were then grouped into categories (e.g. Low to High) so that dropout rates could be compared across levels and the gradient of change shown. Finally, Pearson correlations between each numeric variable and dropout were used to rank the factors by strength of association.

4. Findings

Overall Retention *(Python)*

- 37.1% of learners (16,680) dropped out before reaching B2.
- 35.8% (16,093) reached B2 or above.
- 27.2% (12,227) had not dropped out but were still below B2.
- Most common CEFR level: A2 (24.0%).

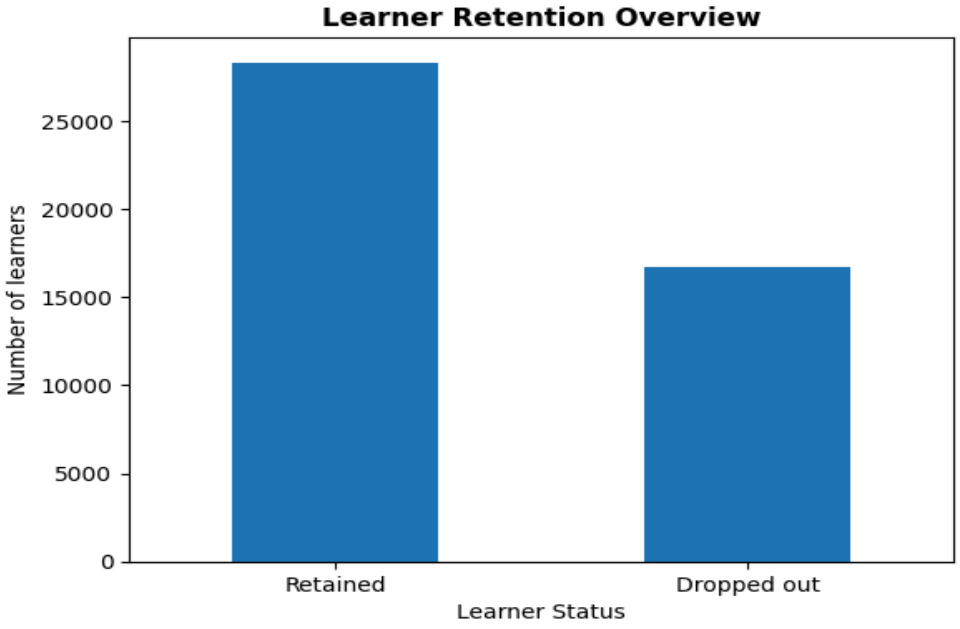


Figure 1. Number of learners who dropped out versus those retained (n = 45,000). "Retained" includes learners who reached B2 and those still below B2.

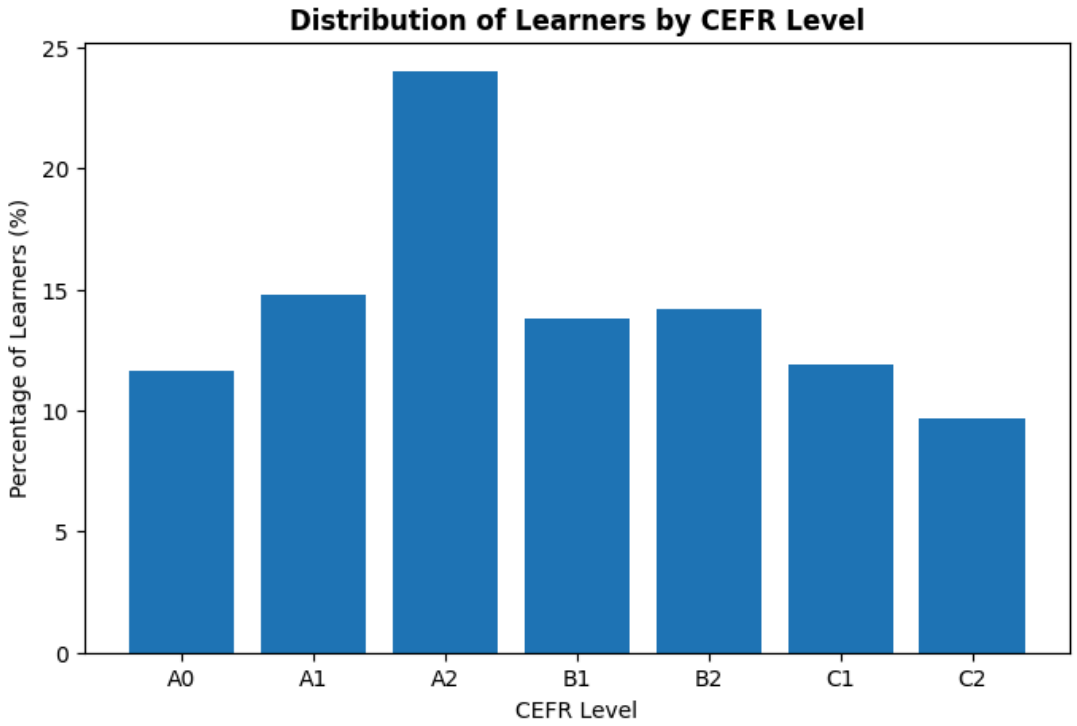


Figure 2. Distribution of Learners by CEFR Level

Motivation

Mean motivation: 5.23 for Retained learners, and 4.58 for Dropout learners.

Motivation Level	Dropout Rate
Low	47.7%
Moderate	38.1%
High	31.1%
Very High	24.6%

The dropout rate steadily fell as motivation increased among learners, with a 23.1 percentage point difference between the Low Motivation and Very High Motivation categories.

Consistency

Mean consistency: 6.67 for Retained learners, and 6.03 for Dropout learners.

Consistency Level	**Dropout Rate**
Low	48.8%
Moderate	42.7%
High	35.9%
Very High	27.8%

The Dropout Rate is highest for the ‘Low Consistency’ group (48.8%) and lowest for the ‘Very High Consistency’ group (27.8%), with a 21 percentage point difference.

FSI Category

FSI Category	Dropout Rate
1	23.4%
2	31.2%
3	42.1%
4	59.7%

- Dropout rates increased as FSI difficulty increased.
- Learners in Category 1 had the lowest dropout rate at 23.4%.
- Category 4 had the highest dropout rate at 59.7%, meaning almost 6 in 10 learners dropped out.
- Overall, the results suggest that more difficult languages were associated with higher dropout rates.

Learner Background

- The number of prior languages made almost no difference: dropout was between 36 to 38% across groups.
- Learners studying a language related to one they knew dropped out slightly less (34% vs 39%).

Study Hours *(Python)*

Learners were split into three equal-sized groups by total study hours.

- ❖ Dropout was 52.6% for low study hours, 39.1% for medium and 18.7% for high
- ❖ The low group's dropout rate was nearly 3 times that of the high group

Comprehensible Input

- Dropout fell from 53% (0 to 130 hours) to 17% (440+ hours)
- Retained learners averaged 64% more comprehensible input hours than those who dropped out

Study Hours and Active Language Use

Dropout rates were compared across combinations of total study hours and active use to examine how these factors were associated with learner retention. Analysing the variables together provided closer insight of whether higher study time was associated with lower dropout rates at different levels of active use, rather than examining each factor independently.
This approach helped identify whether learners with both higher study time and greater active use showed different dropout patterns compared with learners with lower levels of engagement.

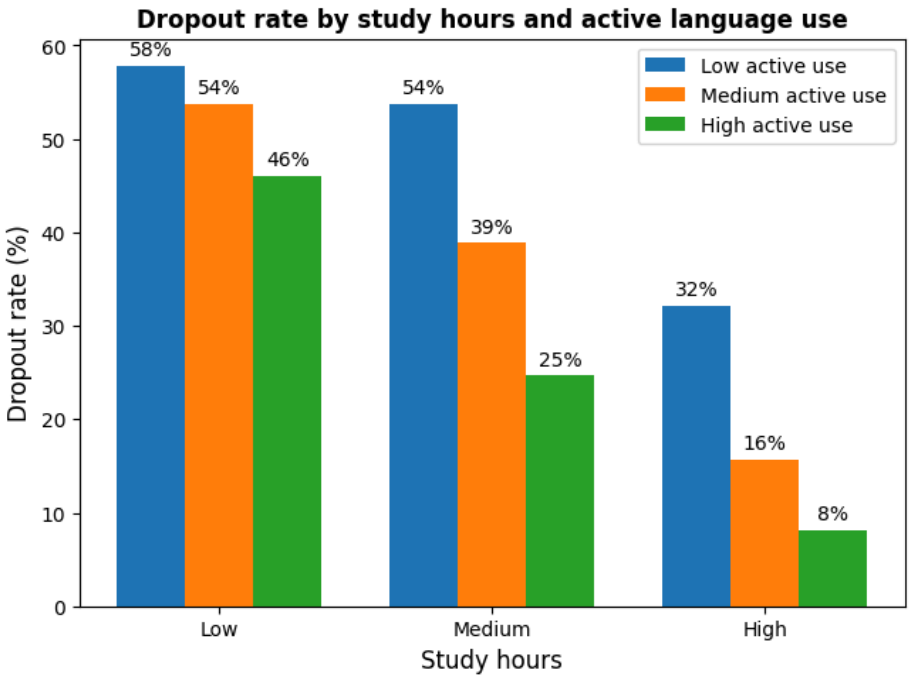


Figure 3. Dropout rate (%) by total study hours and share of study time spent on active language use.

- Dropout ranged from 8.1% (high study, high active use) to 57.8% (low study, low active use).
- Within each study-hours group, higher active use meant lower dropout, by 11.7, 29.0 and 24.1 percentage points for low, medium and high study hours.
- Low study hours with high active use (46.1%) had more dropout than high study hours with low active use (32.2%), suggesting hours had the stronger association.

Study hours, Active Use and Comprehensible Input combined *(SQL)*

- The combination of Low Study Hours, Low Active Use, and Low input had the highest dropout rate of 54.3%.
- The combination of High Study Hours, High Active Use, and High input had the lowest dropout rate of 10.5%.

Ranking the Factors *(Python)*

Pearson correlations between each numeric variable and dropout showed which factors were most strongly associated.

- Strongest associations: total study hours (r = −0.29), comprehensible input (−0.27), and language difficulty (+0.27).
- Active use, motivation and consistency showed weaker but consistent relationships (r ≈ −0.14 to −0.19).
- Language difficulty (fsi_category) is the only variable positively associated with dropout.
- Age and prior languages showed close to none.

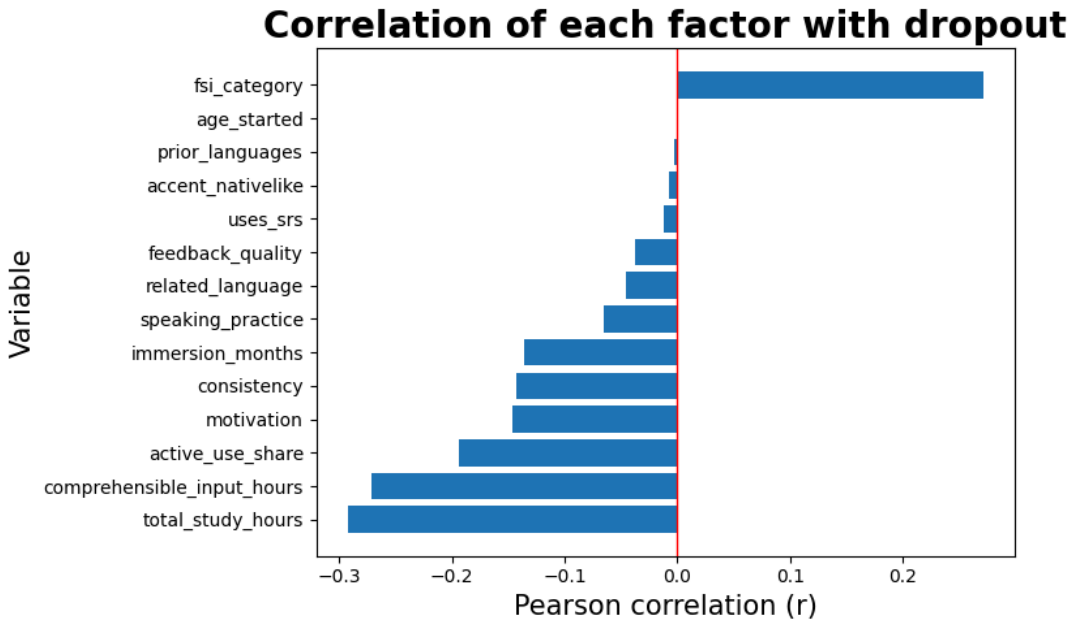


Figure 4. Pearson correlation (r) between each numeric variable and dropout. Negative values indicate lower dropout at higher values of the variable.

Limitations

The dataset studied is synthetic. This is evident as each row of data was artificially generated. This is a weakness as the data may not be an accurate representation of learner behaviour. Therefore, the findings may not be generalisable for real language learners and would need validation using authentic data.

Another weakness is that the measurement of some variables is unclear. For example, motivation, consistency and feedback quality are scored on 0-10 scales, but the dataset does not state how these were measured or at what point in the learner's journey. This is a weakness as, if the scores were self-reported or recorded after learners had already dropped out, they may be affected by biases such as social desirability bias and recall bias. Therefore, the associations between these variables and dropout may not accurately demonstrate which learner behaviours are associated with dropout.

Conclusion

This report analysed the Language Learning to Fluency dataset to identify which learner behaviours and characteristics are associated with dropout. Total study hours, comprehensible input hours and language difficulty showed the strongest associations: dropout was lower among learners with more study time and input, and higher for harder languages. Motivation, consistency and active use showed weaker but steady associations, while age and prior languages showed almost none. Dropout was 10.5% for learners high on study hours, active use and input, versus 54.3% for those low on all three. As the data is synthetic and the analysis observational, these findings show associations only and would require validation using real learner data.